

Grade 2 Ontario Curriculum Data

What Could Happen?

Unit Overview: What Could Happen?

Grade: Grade 2 | **Strand:** Data

Target Expectations: D2.1, D2.2

Learning Goal: Children will use likelihood vocabulary including complementary events, predict outcomes, and test mode predictions across populations.

Lesson	Lesson Title	Learning Goal
Day 1	Likelihood Review	I can use likelihood vocabulary.
Day 2	Complementary Events	I can name complementary events.
Day 3	Spinner Predictions	I can predict spinner outcomes using likelihood.
Day 4	Dice Predictions	I can predict the mode of dice rolls.
Day 5	Mode Prediction Capstone	I can predict mode and test it across groups.

Lesson 1: Likelihood Review

Goal: Events are IMPOSSIBLE, POSSIBLE, or CERTAIN. We predict using likelihood.

Expectation: D2.1

Learning Goal: "I can use likelihood vocabulary."

Materials: M1_likelihood_chart, M7_event_cards, M10_unit_poster

1. Minds On (12 mins): Lucky Reviews Likelihood

Lucky: WILL it snow today? Sometimes! Will the sun come up tomorrow? CERTAIN! Will I fly to the moon? IMPOSSIBLE! Lucky opens: 'NEVER, ALWAYS, or SOMETIMES?' Class: 'I, C, or P!'

Bring children to the carpet. Lucky and Mae puppets. 'Friends — last year you learned likelihood. Today we REVIEW.'

2. Action (25 mins): Likelihood Stations

- Whole group: name 6 events as I/P/C with Canadian context.
- Three stations.
- Stretch at C: justify each likelihood.
- Discuss the 3 categories.
- Circulate to record diagnostic evidence.
- STRATEGY COMPARE: Two children share contrasting events for the same category.

3. Consolidation (13 mins): Lucky's Likelihood Brag

Discuss:

- What's IMPOSSIBLE?
- What's POSSIBLE?
- What's CERTAIN?
- What does PREDICT mean?

Exit: Lucky's refrain — class chants: Lucky calls 'NEVER, ALWAYS, or SOMETIMES?' Class chants 'I, C, or P!' (Impossible, Certain, Possible).

Lesson 2: Complementary Events

Goal: Complementary events: when ONE happens the OTHER doesn't. Heads OR tails. Rain OR no rain.

Expectation: D2.1

Learning Goal: "I can name complementary events."

Materials: M2_complement_cards, M8_complement_chart, M1_likelihood_chart, M10_unit_poster

1. Minds On (12 mins): Mae Pairs Up Complements

Mae: HEADS or TAILS! RAIN or NO RAIN! Either THIS happens or THAT happens — they're COMPLEMENTARY! Mae opens: 'EVENT — and its OPPOSITE! Both are POSSIBLE!'

Bring children to the carpet. Lucky and Mae puppets. 'Friends — yesterday we reviewed likelihood. Today we learn COMPLEMENTARY events.'

2. Action (25 mins): Complement Stations

- Whole group: match 4 events to their complements.
- Three stations.
- Stretch at C: 3 different complement pairs.
- Discuss why 'hot or cold' is NOT strictly complementary.
- Circulate to record formative evidence.
- STRATEGY COMPARE: 2-3 child-generated pairs to the carpet.

3. Consolidation (13 mins): Mae's Complement Brag

Discuss:

- What's a COMPLEMENTARY event?
- Give an example.
- Is 'hot or cold' strictly complementary?
- What's the complement of HEADS?

Exit: Mae's complement refrain — Mae calls 'EVENT — and its OPPOSITE!' Class: 'Both are POSSIBLE!' (3x).

Lesson 3: Spinner Predictions

Goal: A SPINNER with bigger sections is MORE LIKELY to land there.

Expectation: D2.1

Learning Goal: "I can predict spinner outcomes using likelihood."

Materials: M3_spinner_set, M5_prediction_mat, M9_outcome_cards, M10_unit_poster

1. Minds On (12 mins): Lucky Spins the Spinner

Lucky: Look at this spinner — RED is the BIG section, BLUE is small. Predict the spin — most likely RED! Lucky opens: 'BIG section, BIG chance — small section, small chance!'

Bring children to the carpet. Lucky and Mae puppets. 'Friends — yesterday complementary events. Today SPINNERS.'

2. Action (25 mins): Spinner Stations

- Whole group: predict 4 spinners' most likely outcomes.
- Three stations.
- Stretch at C: design a 3-colour spinner with one big section.
- Discuss equal vs unequal spinners.
- Circulate to record formative evidence.
- STRATEGY COMPARE: After Station A, two pairs share HOW they decided which section was 'biggest' — by eye, by counting equal slices, by overlay.

3. Consolidation (13 mins): Lucky's Spinner Brag

Discuss:

- What's a SPINNER?
- What does BIGGER section mean?
- Did your prediction match?
- What if all sections are equal?

Exit: Lucky's spinner refrain — Lucky calls 'BIG section, BIG chance —' Class: 'small section, small chance!' (3x with hand gesture).

Lesson 4: Dice Predictions

Goal: Dice rolls are POSSIBLE 1-6. We predict the MODE after many rolls.

Expectation: D2.2

Learning Goal: "I can predict the mode of dice rolls."

Materials: M4_dice_set, M5_prediction_mat, M9_outcome_cards, M10_unit_poster

1. Minds On (12 mins): Mae Rolls and Tracks

Mae: I'll roll this die 10 times. Predict the MODE — the most common number!
Mae opens: 'Each roll is FRESH — the die FORGETS!'

Bring children to the carpet. Lucky and Mae puppets. 'Friends — yesterday spinners. Today DICE and MODE.'

2. Action (25 mins): Dice Stations

- Whole group: roll die 12 times.
- Three stations.
- Stretch at C: predict if same mode will appear in a 3rd roll.
- Discuss randomness.
- Circulate to record formative evidence.
- STRATEGY COMPARE: After Station A, two pairs share HOW they could PREDICT mode of 6 rolls.

3. Consolidation (13 mins): Mae's Dice Brag

Discuss:

- What's MODE?
- Are all 6 numbers equally likely?
- Did mode change between rolls?
- Why?
- Mae's preview: tomorrow we test mode across MANY trials (Lesson 5 — D2.2 bridge).

Exit: Mae's fresh-die refrain — Mae calls 'Each roll is FRESH —' Class: 'the die FORGETS!' (3x with open-palm wipe).

Lesson 5: Mode Prediction Capstone

Goal: Detectives predict the MODE for one group, then test in ANOTHER group. Same mode? Different?

Expectation: D2.1, D2.2

Learning Goal: "I can predict mode and test it across groups."

Materials: M1_likelihood_chart, M2_complement_cards, M3_spinner_set, M4_dice_set

1. Minds On (12 mins): Lucky and Mae Plan the Capstone

Lucky: Today's CASE — predict the MODE in one group, then TEST in another. Same? Different? Lucky+Mae open: 'Same die, different ROLLS! Different MODES! PREDICTION uses the DATA!'

Bring children to the carpet. Lucky and Mae puppets in lab coats. 'Friends — today our CAPSTONE. We use ALL we know: likelihood, complementary, spinner, dice, MODE prediction.'

2. Action (25 mins): The Mode Prediction Lab

- Hand out templates.
- Station 1 (LIKELIHOOD): sort 6 events I/P/C.
- Station 2 (COMPLEMENT): match 4 pairs.
- Stations 3-4 (SPINNER + DICE): predict + test.
- Station 5 (SURVEY): predict mode + test.
- COMPARE TWO PREDICTIONS: Present 'Lin says SAME mode because both classes used the same die'; 'Sam says DIFFERENT mode because each set is fresh.' Class debates with data — both...

3. Consolidation (13 mins): Lucky's Mode Brag

Discuss:

- What's COMPLEMENTARY?
- What was your dice mode?
- Did your prediction match the test?
- Why might mode differ between groups?

Exit: Closing capstone refrain — Lucky+Mae call-and-response: Lucky: 'Same die, different ROLLS!' Mae: 'Different MODES!' Class: 'PREDICTION uses the DATA!' (3x).

Name: _____

Date: _____

Lucky's Likelihood Review

Learning Goal: I can use likelihood vocabulary.

Instructions for the Student:

1. Read each event.
2. Write a prediction about TODAY using a likelihood word.
3. Write 1 IMPOSSIBLE, 1 POSSIBLE, 1 CERTAIN event.

Sort by Likelihood

Sort 6 events into the 3 boxes.

Name: _____

Date: _____

Mae's Complement Match

Learning Goal: I can name complementary events.

Instructions for the Student:

1. Draw a line from each event to its complement.
2. Sort 6 pairs as COMPLEMENTARY or NOT.
3. Write 3 of your own complement pairs.

Sort and Explain

Sort with reasons.

Name: _____

Date: _____

Lucky's Spinner Lab

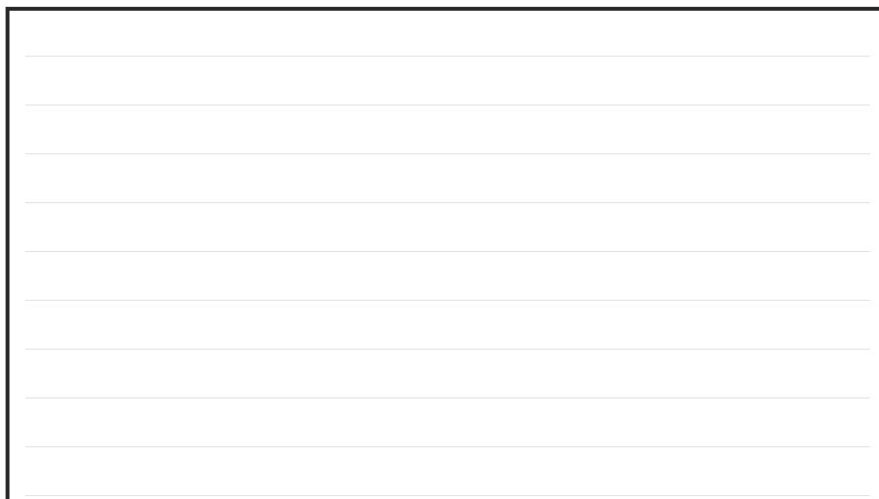
Learning Goal: I can predict spinner outcomes using likelihood.

Instructions for the Student:

1. Look at each spinner.
2. Spin a real spinner 5 times.
3. Design a spinner with 3 colours where ONE is most likely.

Part 3: Design your own.

Design a spinner with 3 colours where ONE is most likely.

A large rectangular box with a black border, containing ten horizontal grey lines for drawing a spinner.

Name: _____

Date: _____

Mae's Dice Lab

Learning Goal: I can predict the mode of dice rolls.

Instructions for the Student:

1. Roll a die 12 times.
2. Roll AGAIN 12 times.
3. PREDICT the mode of a 3rd 12-roll.

Part 2: Second roll.

Roll AGAIN 12 times. Find the new mode. Compare to first.

Name: _____

Date: _____

Lucky and Mae's Mode Lab

Learning Goal: I can predict mode and test it across groups.

Instructions for the Student:

1. Sort 4 events I/P/C.
2. Predict + test a spinner.
3. Survey 5 classmates.

Likelihood Capstone

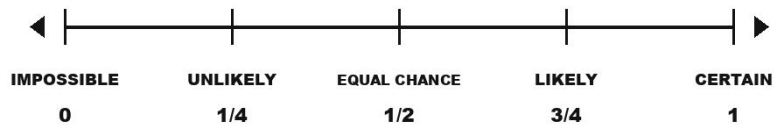
Capstone: likelihood scale.

Likelihood Chart

(M1_likelihood_chart)

Anchor chart of impossible/possible/certain.

Likelihood Scale



PRINT SPECS

Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 1 copy.
2. Laminate.
3. Tape to front wall before Lesson 1.

USE IN CLASS

Quantity: 1 chart
Lessons: 1, 2, 5
Prep: ~10 min

Complement Cards

(M2_complement_cards)

Set of 12 paired cards showing complementary events.

Complementary Events

- When **ONE** happens, the **OTHER** doesn't.
- Heads ↔ Tails. Rain ↔ No Rain.
- Win ↔ Lose. Day ↔ Night.
- Find **ONE**, get the **OTHER** for free.

PRINT SPECS

Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 10 sets.
2. Laminate.
3. Cut into 12 cards per set.
4. Store in labelled bag.

USE IN CLASS

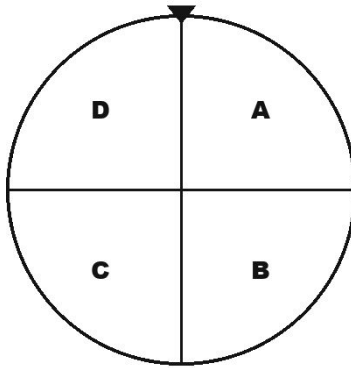
Quantity: 10 sets
Lessons: 2, 5
Prep: ~15 min

Paper Spinner Set

(M3_spinner_set)

Set of 6 paper spinners with varied section sizes.

Spinner



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut out spinners.

4. Attach paper-clip pointers with brads.

USE IN CLASS

Quantity: 10 sets

Lessons: 3, 5

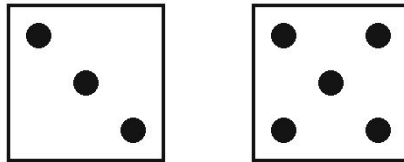
Prep: ~30 min

Paper Dice Set

(M4_dice_set)

Set of 6 foldable paper dice.

Dice



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: no

TEACHER PREP

1. Print 6 sets.

2. Cut and fold each die.

3. Tape tabs to assemble.

4. Store in labelled box per pair.

USE IN CLASS

Quantity: 10 sets

Lessons: 4, 5

Prep: ~45 min

Prediction Recording Mat

(M5_prediction_mat)

Letter-size mat for recording predictions and outcomes.

Outcome Table

Trial	Outcome	Tally	Probability

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 12 mats.

2. Laminate for dry-erase.

3. Distribute one per pair.

USE IN CLASS

Quantity: 10-12 mats

Lessons: 3, 4, 5

Prep: ~10 min

Mode Prediction Template

(M6_capstone_template)

Letter-size capstone template.

Mode Prediction Template	
1	Likelihood _____
2	Complement _____
3	Spinner _____
4	Dice _____

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

TEACHER PREP

1. Print 25 templates.

2. Stack near teacher table.

3. No lamination.

USE IN CLASS

Quantity: 25 templates

Lessons: 5

Prep: ~5 min

Event Cards

(M7_event_cards)

Set of 16 event cards for likelihood review.

Event Cards

The sun rises tomorrow.	It will snow today.
I will pick a red marble.	I will roll a 7.
Coin lands on heads.	Tomorrow is Wednesday.
The ice will melt in the sun.	Dog will bark at noon.

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 16 cards per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

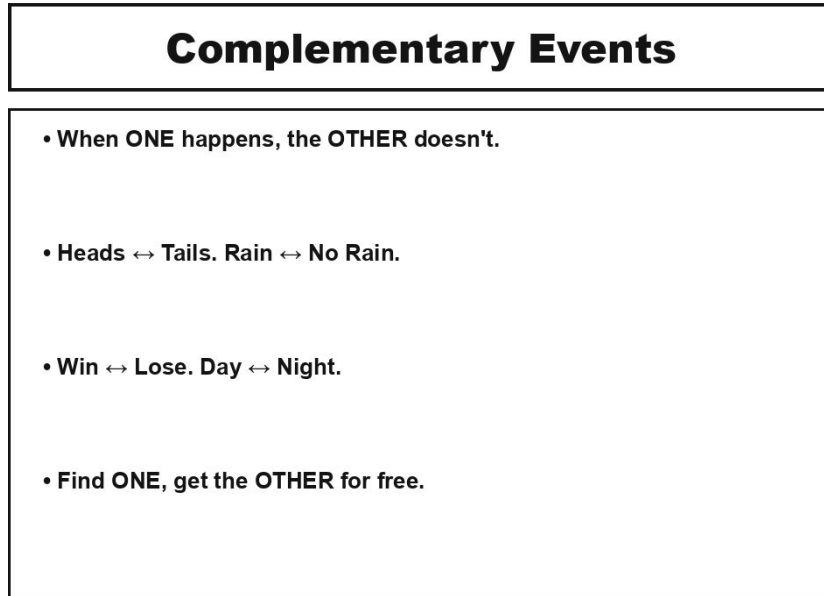
Lessons: 1, 2, 5

Prep: ~15 min

Complementary Events Chart

(M8_complement_chart)

Anchor chart explaining complementary events.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape next to likelihood chart.

USE IN CLASS

Quantity: 1 chart

Lessons: 2, 5

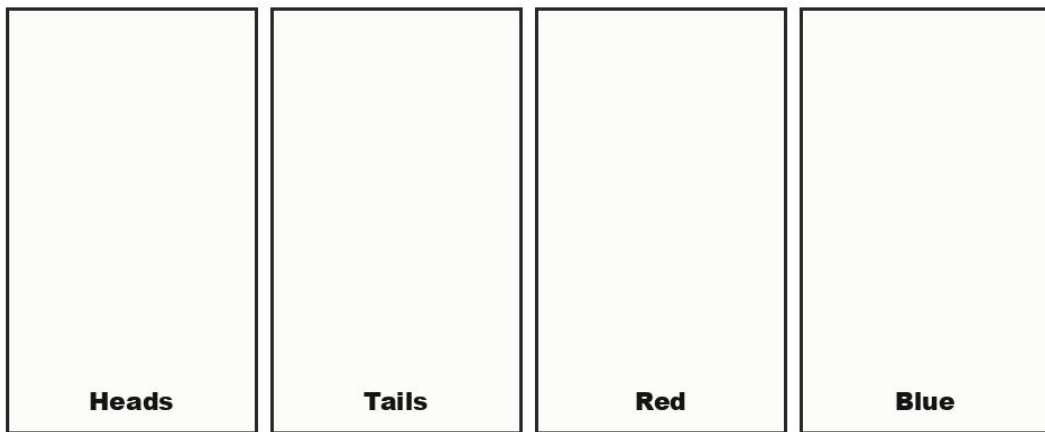
Prep: ~10 min

Outcome Cards

(M9_outcome_cards)

Set of 12 outcome icons (heads/tails, sun/rain).

Outcome Cards



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 12 cards per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

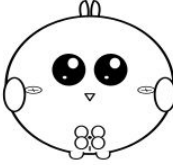
Lessons: 3, 4, 5

Prep: ~15 min

What Could Happen? Poster

(M10_unit_poster)

Anchor poster of Lucky and Mae.

What Could Happen? Poster		
 Lucky Lou the Loon	 Maybe Mae the Cat	Likelihood
Complement Pair	Spinner	Dice

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy on letter or 11x17.

2. Laminate.

3. Tape to front wall on Day 1.

USE IN CLASS

Quantity: 1 poster

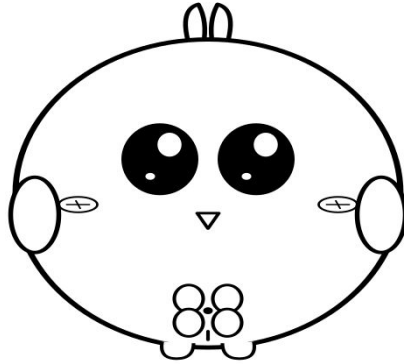
Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

Lucky Lou Puppet

(char_lucky_puppet)

Paper puppet of Lucky for teacher use in all 5 lessons.



Lucky Lou the Loon

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Cut around silhouette + tab.

3. Laminate.

4. Glue popsicle stick.

USE IN CLASS

Quantity: 1 puppet

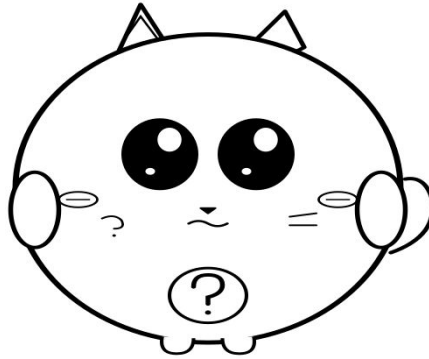
Lessons: 1, 2, 3, 4, 5

Prep: ~12 min

Maybe Mae Puppet

(char_maybe_puppet)

Paper puppet of Mae for teacher use in Lessons 2-5.



Maybe Mae the Cat

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Cut around silhouette + tab.

3. Laminate.

4. Glue popsicle stick.

USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 3, 4, 5

Prep: ~12 min

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
D2.1 use mathematical language, including the terms “impossible”, “possible”, and “certain”, to describe the likelihood of complementary events happening, and use that likelihood to make predictions and informed decisions	Defaults to POSSIBLE for everything. Cannot identify complement pairs.	Names I/P/C with prompts. Identifies complement pairs in obvious cases.	Independently uses likelihood vocabulary AND identifies complementary events. Predicts spinner outcomes correctly.	Distinguishes strict complements from non-complements (e.g., hot/cold has warm in between).
D2.2 make and test predictions about the likelihood that the mode(s) of a data set from one population will be the same for data collected from a different population	Cannot predict mode. Predicts same mode regardless of group.	Predicts mode for one group with prompts. Tests in second group.	Independently predicts mode in one group and accepts that another group's mode may differ.	Reasons about WHY modes differ across groups (different favourites).

Junior Likelihood Detective Certificate (G2)

This certificate is proudly presented to:

For successfully completing the 5-Day What Could Happen? unit and demonstrating the skills of a G2 Junior Likelihood Detective. Lucky Lou the Loon congratulates the student.

By completing this unit, this student has demonstrated:

- Using likelihood vocabulary
- Naming complementary events
- Predicting spinner outcomes
- Finding mode and predicting across groups

Marketplace Listing

A 5-day Grade 2 unit on likelihood framed around Lucky Lou the Loon and Maybe Mae the Cat.

Price: \$8.99 CAD

Pages: ~42

Classroom time: 250 min

Teacher prep: ~200 min

Top tags: Grade 2, math, data, likelihood, probability, complementary, spinner, dice

What's included:

- Unit blueprint (1 file)
- 5 lesson plans (5 files)
- 5 worksheets (5 files)
- Manipulatives plan (1 file)
- Formative + reflection (1 file)
- Assessment suite (1 file)